17 Heidi wants to make some biscuits using this recipe.

Makes 12 biscuits
125 g butter
200 g flour
50 g sugar

Heidi thinks that she has,

500 g butter
700 g flour
250 g sugar

Assuming that these weights are correct,

(a) work out the greatest number of biscuits Heidi can make.
You must show all your working.

Butter allows 500/125 = 4 batches → 48 biscuits.

Flour allows 700/200 = 3.5 batches → 42 biscuits.

Sugar allows 250/50 = 5 batches → 60 biscuits.

Flour limits Heidi to 42 biscuits.

4 marks total
One ingredient
All three limits
Multiply by 12
42 biscuits

(4)

Heidi is wrong.
She has more than 250 g of sugar.

(b) Does this affect the greatest number of biscuits Heidi can make?
Give a reason for your answer.

1 mark
Flour limits quantity
No change

No. Flour is still the limiting ingredient;

extra sugar cannot increase the 42-biscuit maximum.

(1)

(Total for Question 17 is 5 marks)

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